IEEE VTS | IEEE Vehicular Technology Society

IEEE

IEEE VTC Spring 2027 · Workshop 7

Synergizing Digital Twins and Pervasive Intelligence for Next-Generation IoV

The 4th Workshop · Call for Papers

Next-generation Internet of Things (IoT) and Internet of Vehicles (IoV) systems are becoming increasingly autonomous, interconnected, and intelligence-driven. Their communication, computing, and control infrastructures must sustain high mobility, dense connectivity, heterogeneous services, and stringent latency, reliability, and safety requirements. Digital twins (DTs) offer continuously updated virtual representations of vehicles, networks, and transportation environments, while pervasive artificial intelligence (AI) distributes learning, inference, prediction, and decision-making across vehicles, roadside units, edge servers, and cloud platforms. Together, they can form a bidirectional physical-virtual intelligence loop: DTs support data-driven modeling, experimentation, and optimization, while AI improves state estimation, prediction, adaptation, and closed-loop control. Key barriers remain. Rapid mobility, time-varying channels, incomplete observations, and evolving topologies can delay updates, desynchronize twins, and widen the physical-virtual domain gap. At the same time, large-scale twin maintenance and distributed AI increase communication and computing load, which can reduce information freshness and decision quality.

This half-day workshop will convene researchers and practitioners to advance scalable, trustworthy, and resource-aware DT-AI systems for next-generation IoT/IoV. Topics include adaptive synchronization, domain adaptation, edge-cloud collaboration, task partitioning, cooperative processing, computation offloading, distributed inference, joint communication-computation orchestration, and validation through prototypes, testbeds, and field trials.

Topics of Interest (including but not limited to)

- DT architectures and modeling techniques for IoV systems
- AI-enhanced digital twins for prediction, optimization, and autonomous control
- Physical-virtual synchronization, DT fidelity, and state reconstruction
- Domain adaptation, continual learning, and cross-domain knowledge transfer
- Pervasive intelligence in self-organizing and autonomous vehicular systems
- Edge-cloud collaboration for DT-enabled intelligence
- Distributed and cooperative learning for DT-AI integration
- Resource management, routing, scheduling, and network slicing enabled by DT-AI
- Security, privacy, and trust in DT-AI-enabled IoT/IoV ecosystems
- Interoperability and standardization of DT and AI technologies for IoT/IoV and 6G
- Big data management and real-time analytics for large-scale digital twins
- DT-AI-enabled applications in intelligent transportation and smart infrastructure
- UAV/drone-assisted IoT/IoV sensing and monitoring using DT and AI
- Experimental platforms, testbeds, and field trials for DT-AI-enabled IoT/IoV
- Cross-domain innovations enabled by DT-AI synergy in smart mobility and beyond

SUBMISSION GUIDELINES

Papers must be formatted in the standard IEEE two-column format that is used by the VTC-Spring 2027 main conference. 5-page paper (without overlength charge) and up to 2 additional pages are allowed with the purchase of additional page charges in the amount of $100 USD per additional page at the time of registration and final paper submission.

IMPORTANT DATES

Paper Submission Deadline:	**20 Dec. 2026**
Paper Acceptance Notification:	**1 Feb. 2027**
Final Paper Submission:	**TBD**

Organizing Committee

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Qiang (John) Ye, University of Calgary, Canada
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